Hector Ovalle

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EDUCATION

Georgia Institute of Technology **May 2026 - December 2028**
Bachelor of Science in Electrical Engineering **GPA: 4.0**
Relevant Coursework: Digital System Design, Circuit Analysis, Signal Processing, Engineering Software Design
Professional Organizations: SiliconJackets, RoboJackets, Society of Hispanic Professional Engineers

EXPERIENCE

Augusta University
Computer Science Undergraduate Course Assistant **August 2025 - December 2025**

- Embedded in multiple course sections to assist students in real time, answering conceptual and coding-related questions to improve understanding and workflow.
- Guided students in using GitHub for version control and collaborative development, increasing code accessibility and ease of iteration.
- Contributed to a 34% improvement in overall student performance compared to previous semesters.

PROJECTS

32-bit RISC-V CPU - Verilog/SystemsVerilog **July 2026 - August 2026**

- Designed and implemented a 32-bit RISC-V CPU with instruction fetch, register-file interfacing, instruction decoding, and ALU logic.
- Implemented arithmetic, immediate, shift, load/store, branch, and halt instructions, including sign-extension for RISC-V immediate formats.
- Integrated instruction/data SRAM interfaces and verified processor behavior through simulation and debugging in a Linux development environment.

Daisy-Chained Sensor Suite PCB | KiCad, RP2040, I²C, SPI **August 2026 - September 2026**

- Designed an RP2040-based daisy-chainable sensor PCB integrating six I²C peripherals, SPI communication interfaces, and Micro-USB connectivity for power delivery and firmware programming.
- Developed detailed circuit schematics and multilayer PCB layouts in KiCad, incorporating power distribution, signal routing, component placement, and communication interfaces for modular sensor integration.
- Researched component datasheets and documented hardware specifications in a Hardware Reference Manual (HRM), ensuring design compatibility with JLCPCB manufacturing constraints and supporting future firmware development.

Magnetic Induction and Electromagnetic Coil **August 2025 - December 2025**

- Designed and constructed an electromagnetic coil system to investigate magnetic induction and induced voltage through hands-on experimentation.
- Utilized an oscilloscope to measure and analyze voltage waveforms, evaluating electromagnetic behavior and validating theoretical principles.

TECHNICAL SKILLS

Programming: Python, C#, C++, Javascript, MATLAB, HTML/CSS
Hardware: PCB Design, Arduino, Oscilloscope, AC/DC power supplies,DMM, Hardware debugging
Software: KiCAD, Multisim, VS Code, Anaconda, OpenCV, GitHub, Microsoft Suite, Arduino IDE, Eclipse
HDL: Verilog/SystemVerilog